

Negin Maddah

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Education

Northeastern University, Boston, MA. Ph.D., Industrial Engineering. Jan 2022–May 2027

- Dissertation: “Governing Information Systems in the Generative AI Era: Centralized Control of Decentralized Participation and Workflow” (in progress; advisor: Babak Heydari).
- Graduate GPA: 3.9/4 | Concurrent Graduate Certificate in Data Analytics Engineering.

K. N. Toosi University of Technology, Tehran, Iran. M.Sc., Industrial Engineering. Sep 2017–Jan 2020

Alzahra University, Tehran, Iran. B.Sc., Industrial Engineering. Sep 2013–Sep 2017

Research Interests

AI adoption and the organization of work | digital platforms, collaboration, and knowledge production | information systems governance, security and compliance controls, and process mining | sociotechnical systems modeling | network resilience | trustworthy analytics and data quality.

Research Appointments

Northeastern University, MAGICS Lab. Graduate Research/Teaching Assistant. Jan 2022–May 2027

Harvard Medical School, NEZAMI Lab. AI Researcher Intern. May 2025–Dec 2025

Peer-Reviewed Journal Articles

1. Maddah, N., and B. Heydari. 2024. “**Platform-Driven Collaboration Patterns: Structural Evolution Over Time and Scale.**” *IEEE Transactions on Computational Social Systems* 11(6): 7814–7829. DOI.
 - Recognition. Plenary talk (one of **20/700+** submissions), IC2S2 (International Conference on Computational Social Science) 2024, University of Pennsylvania.
2. Maddah, N., and B. Heydari. 2024. “**Building Back Better: Modeling Decentralized Recovery in Sociotechnical Systems Using Strategic Network Dynamics.**” *Reliability Engineering & System Safety* 246: 110085. DOI.
 - Models post-disruption recovery as a decentralized, bottom-up process, extending strategic network formation to heterogeneous agents whose reconnection choices reshape structure over time. Quantifies recovery that no central authority directs, which is the condition continuity and incident-recovery planning actually face.
 - Recognition. Best Presentation Award, CESUN (Council of Engineering Systems Universities) 2025.
3. Maddah, N., and E. Roghanian. 2023. “**Data-Driven Performance Management of Business Units Using Process Mining and DEA: Case Study of an Iranian Chain Store.**” *International Journal of Productivity and Performance Management* 72(2): 550–575. DOI.
 - Based on master’s thesis research.
 - Industry impact. The case-study firm extended a business analyst offer following the performance-management framework proposed in the study.

Manuscripts under Revision

4. Heydari, B., and N. Maddah. “**Public Knowledge Production in the Generative AI Innovation System: Selective Reconfiguration of Stack Overflow after ChatGPT.**” *Technological Forecasting and Social Change*, revise and resubmit.
 - Examines changes in contributor renewal, demand composition, and answer access following ChatGPT’s release.

5. **Maddah, N.**, A. Ramezani, Q. Jin, J. Wollborn, A. Itoh, J. B. Rabb, F. Mazhude, R. S. Kramer, D. B. Sawyer, R. L. Winslow, and F. R. Nezami. “**Generative AI Imputation to Preserve Data Fidelity and Outcome Prediction.**” *npj Digital Medicine*, revise and resubmit.
 - Benchmarks classical and generative approaches to missing-data reconstruction across mixed data types in two electronic health record systems, distinguishing how faithfully values are reconstructed from how much they improve outcome prediction.
6. Ramezani, A., **N. Maddah** (co-first author), M. Heine, A. Homaei, L. Simeth, L. Mazuera, A. Osho, J. D. Muehlschlegel, J. Wollborn, and F. R. Nezami. “**BeatAI: BiomEtrics for Real-Time Atrial Arrhythmia Tracking Using Artificial Intelligence.**” *BMC Cardiovascular Disorders*, revise and resubmit.
 - Develops daily risk classification and hour-ahead forecasting using transformers.
 - **Recognition.** Selected spotlight presentation, MIT–MGB AI-Cures 2025.

Working Papers

1. **Maddah, N.**, and B. Heydari. “**Individual AI-Assistance and Workflow Change in Open-Source Production: Evidence from AI-Agent Adoption.**” Working paper.
 - Separates an individual AI assistance channel from a workflow channel in AI-agent adoption on GitHub.
 - Identifies how each channel changes what contributors produce and how repositories convert contributions into accepted output.
 - Traces the two jointly to explain the resulting redistribution of code authoring and reviewing, with heterogeneity by prior expertise and role.
 - Built on an event log of **33.1 million events** across **65** GitHub repositories, Jan 2021–Jun 2026.
2. **Maddah, N.**, and B. Heydari. “**LLM for Centralized Process Compliance Checking in Decentralized Co-Creation.**” Working paper (target: *Information Systems Research* or *Organization Science*).
 - Recovers procedural and technical demand that event logs timestamp but do not encode.
 - Uses a blinded, evidence-citing language-model instrument over **3,834** GitHub code-review cases, checked by an independent second LLM scorer.
 - Audits four change-approval controls (code landing unreviewed or over a rejection, reversed verdicts, stale approvals) across **1,495** sampled changes in **31** GitHub projects.
 - Speaks to security and compliance governance, where a violation count alone cannot separate a control failure from a lagging audit trail.
3. **Maddah, N.**, and B. Heydari. “**Technology Shock or Network Feedback? A Behavior-Calibrated Markov Model.**”
 - In progress. Learns a covariate-dependent model of contributor state transitions from pre-release behavior.
 - Embeds the learned transitions in a multi-agent simulation over the interaction network.
 - Compares simulated counterfactuals with observed trajectories to isolate the direct generative AI shock from the endogenous amplification network feedback produces.

Conference Presentations and Participation

Presented research at or participated in international conferences including CESUN (Council of Engineering Systems Universities), 2025 and earlier meetings; IC2S2 (International Conference on Computational Social Science), 2024; the INFORMS Annual Meeting, 2024; the Network Science Conference (NetSci), 2026; and the Wiki Workshop, 2024.

- **Selected participant.** FutureBAProf 2024, Tippie College of Business, University of Iowa, a competitive workshop for prospective business analytics faculty.
- Paper-level presentation awards and selections are listed with their respective publications above.

Professional Experience

Behfa Consulting Group | Golrang Industrial Group, Iran. Systems & Processes Analyst. 2018–2021

- Managed data and process mining projects using enterprise-system logs to diagnose bottlenecks, assess process compliance, and inform system improvements.
- Gathered software requirements through stakeholder interviews and system benchmarking; worked with developers on feature testing, validation, deployment, and user adoption.

Teaching and Course Development

Economic Decision-Making. Graduate Teaching Assistant; technical-session instructor. 2023–present
Engineering Management Program, College of Engineering, Northeastern University, Boston, MA.

- Teach collaborative, hands-on technical sessions in online and in-person sections; topics include data simulation, quantitative decision analysis, and financial risk analysis.
- Teaching experience across cohorts totaling approximately **250** graduate students.
- Run exam-preparation sessions and office hours; grade assignments and projects; contribute to exam design and revision.

Economic Decision-Making: online course development. Course co-designer and on-camera instructor. 2025
Northeastern University/Coursera; EDGE Studio, Boston, MA.

- Co-designed the Coursera version for Engineering Management master's students.
- Scripted instructional video modules, co-designed and coordinated animations, and recorded course instruction on camera.

Analytics workshops. Data and process mining workshop developer and instructor. 2021
Sharif University of Technology and Behfa Consulting Group clients, Tehran, Iran.

Honors and Awards

Best Presentation Award, CESUN (Council of Engineering Systems Universities). 2025

Research Merit Award, MIE Department, College of Engineering, Northeastern University. 2024

Academic Service

Journal reviewing: reviewer for journals including *Journal of Computing and Information Science in Engineering*; *Data Mining and Knowledge Discovery*; *Scientific Reports*; *Humanities and Social Sciences Communications*; *Future Business Journal*; *Knowledge and Information Systems*.

Conference reviewing: IC2S2, 2025; Wiki Workshop, 2025.

Methods and Technical Preparation

Methods: Statistical modeling and inference | generative AI (adversarial and autoencoder generative models, large language models) | machine learning (gradient boosting, random forests, transformer and recurrent architectures) | quasi-experimental causal inference (difference-in-differences, event studies, synthetic control, regression discontinuity in time) | network analysis and strategic network formation | agent-based simulation | time-series modeling and forecasting | process mining (process discovery, conformance checking) | text mining and natural language processing.

Tools: Python (pandas, NumPy, PyTorch, TensorFlow, Hugging Face Transformers, NLTK, NetworkX) | R | SQL | Git.

Certificates: Applied Social Network Analysis in Python (2024) | Business Analyst with Process Mining, Celonis (2021).